ASBAT Education Consult

Explore! Discover! Understand! Grow!

Science Explains the World Around Us!

P.5 SCIENCE LESSON PLAN

TERM 2



LIVING THINGS

Understand the characteristics, life processes and classification of living things and their habitats.



MATERIALS

Explore properties of materials and how they are used in our daily lives.



ENERGY

Learn about sources of energy, forms of energy and how energy is used and conserved.



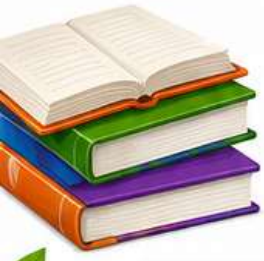
EARTH & SPACE

Discover the Earth, moon, sun and other objects in space and how they affect our lives.



SCIENCE IN OUR WORLD

Apply scientific knowledge and skills to solve problems and make informed decisions.



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12-WEEK TOPICAL BREAKDOWN – P.5 SCIENCE

TERM II

Week	Period	Topic	Sub-topic	Content	Reference
1	1	**COMPONENTS OF THE ENVIRONMENT (SOIL)**	Soil formation and types	Soil – the top most layer of the earth's surface. Formation by weathering and decomposition. Types of soil – loam, sand, clay – particle size, arrangement.	pp. 81-83
	2		Characteristics and uses of soil types	Detailed characteristics of loam, sand, clay (drainage, capillarity, humus content, aeration). Uses of each soil type – crop growing, building, ceramics, glass making.	pp. 83-84
	3		Soil drainage	Drainage – downward movement of water in soil. Experiment comparing drainage in sand, loam and clay. Results and explanation.	pp. 84-85
	4		Soil capillarity	Capillarity – upward movement of water in soil. Experiment comparing capillarity in different soils. Results and explanation.	pp. 85-86
	5		Components of soil	Humus, air, water, living organisms, rock particles, dissolved mineral salts. Importance of each component.	pp. 86-88
2	1		Experiments on soil components	Experiments to show soil contains air (bubbles when placed in water) and water (droplets on heating). Importance of organisms and humus.	pp. 86-88
	2		Soil profile	Soil profile – arrangement of layers (top soil, sub soil, parent rock). Characteristics of each layer. Places where profile is clearly seen.	p. 88
	3		Soil erosion – meaning, agents and causes	Soil erosion – removal of top soil by agents (water, wind, animals). Causes – bush burning, deforestation, overgrazing, overstocking, monocropping.	pp. 89-90
	4		Types of soil erosion and effects	Splash, sheet, rill, gully, river bank, wind erosion. Effects – soil exhaustion, poor yields, silting (deposition in water bodies).	pp. 90-91
	5		Control of soil erosion	Methods – mulching, crop rotation, terracing, contour ploughing, strip cropping,	pp. 91-93

				afforestation, re-afforestation, bush fallowing, agro-forestry. How each controls erosion.	
3	1		Soil fertility and organic manure	Soil fertility – ability of soil to support plant growth. Organic manure – compost, farmyard, green manure. Types of natural fertilizers.	pp. 93-95
	2		Making compost manure	Pit method and heap method – materials used, steps, importance of adding top soil, water, farmyard manure. Turning materials; using a stick to check decomposition.	pp. 96-97
	3		Advantages and disadvantages of organic manure	Advantages – rich in humus, improves soil structure, cheap, stays long, does not pollute. Disadvantages – tiresome, smell, bulky, takes long.	pp. 97-98
	4		Artificial fertilizers	Straight fertilizers (SSP, DSP, TSP) and compound fertilizers (NPK, DAP, CAN). Advantages – quick, high nutrient, easy to apply. Disadvantages – expensive, can poison, pollute water. Methods of application.	pp. 98-99
	5		Soil pollutants and soil barriers	Soil pollutants – polythene, plastics, metal scraps, chemicals, waste oil – effects on soil. Soil barriers – grass, tree roots, terraces, gabions – how they prevent erosion.	pp. 99-100
4	1	**MATTER AND ENERGY**	Energy – types and forms	Energy – ability to do work. Potential energy (at rest) and kinetic energy (in motion). Pendulum illustration. Forms – heat, light, sound, electrical, chemical, solar, magnetic.	pp. 101-103
	2		Matter and its states	Matter – anything that occupies space and has weight. States – solids, liquids, gases. Molecular arrangement, properties, examples. Cohesion and adhesion force.	pp. 103-105
	3		Changes in state and effects of heat	Melting, freezing, evaporation, condensation, sublimation, deposition. Sublimates (mothballs, iodine, solid CO_2). Effects of heat – expansion and contraction.	pp. 105-106

	4		Expansion in solids – ball-and-ring experiment	Experiment with metallic ball and ring – ball passes when cold, fails when heated, passes after cooling. Solids expand when heated and contract when cooled.	pp. 106-107
	5		Bimetallic strip and applications	Bimetallic strip (copper and iron) – copper expands faster, strip bends. Uses – thermostat switches in flat irons, kettles, fridges. Railway line gaps and loose electric wires.	pp. 107-108
5	1		Expansion in liquids	Experiment – flask with coloured water in hot water; first fall (flask expands) then rise (water expands). Application – space left in soda bottles.	pp. 108-109
	2		Expansion in gases	Experiment – balloon on bottle placed in hot water, balloon inflates. Balloon bursts in hot sunshine. Effects of heat on liquids and gases. Water freezing – volume increases, density decreases.	pp. 109-110
	3		Heat transfer – conduction	Conduction – transfer through solids. Wax-on-nail experiment. Conductors (metals) and insulators (plastic, wood, paper). Uses. Water as poor conductor – experiment.	pp. 110-112
	4		Heat transfer – convection	Convection – transfer through liquids and gases. Potassium permanganate experiment (liquid). Smoke box experiment (gas). Convection currents. Land breeze, sea breeze. Ventilation.	pp. 112-113
	5		Heat transfer – radiation and thermos flask	Radiation – heat travel through vacuum. Absorbers (dark/dull) and reflectors (shiny). Thermos flask – parts (cork, vacuum, double silvered walls, metal case) and their functions.	pp. 113-114, 111
6	1		Temperature and thermometers	Temperature – degree of hotness/coldness. Celsius and Fahrenheit scales. Thermometers – clinical (scale 34°C–42°C, kink), Six's (maximum-minimum), wall, ordinary. Mercury vs alcohol.	pp. 112-114
	2		Temperature scale conversion	Formulas: $°F = (°C \times 9/5) + 32$; $°C = (°F - 32) \times 5/9$.	pp. 114-115

				Worked examples and learner practice.	
	3		Burning – candle flame zones	Meaning of burning. Candle flame zones – blue, yellow, central, pale blue – characteristics. Experiment – candle under jar, flame goes out; water rises to fill space.	pp. 115-117
	4		Rusting	Rusting – chemical reaction needing water and oxygen. Experiment – iron nails in different conditions (water+air, dry air, boiled water+oil). Prevention – painting, oiling, galvanising, chromium plating.	pp. 117-118
	5		Mixtures and separation methods	Mixture – combination of substances. Methods – decantation, filtration, evaporation to dryness, distillation, magnetic separation, winnowing, hand picking, separating funnel.	pp. 119-123
7	1		Separating sand and salt; miscible and immiscible liquids	Steps to separate sand and salt – dissolve, filter, evaporate. Miscible liquids (mix) and immiscible liquids (do not mix).	pp. 119-123
	2		Solutions – solute, solvent, saturated solutions	Solute, solvent, solution. Saturated solution (cannot dissolve more on stirring). Super-saturated (cannot dissolve even on heating). Suspensions. Water as universal solvent.	pp. 123-125
	3		Work – definition and calculation	Work = Force × Distance. Units – Joules, Newtons, metres. Calculation examples – a man pushes 60 N through 4 m = 240 J. Practice problems.	pp. 125-126
	4		Friction – types, advantages, disadvantages	Friction – force opposing motion. Types – static, sliding, viscosity. Advantages (walking, writing, braking). Disadvantages (noise, heat, wear).	pp. 126-127
	5		Reducing and increasing friction; diffusion and dilution	Reducing friction – oiling, greasing, ball bearings, rollers. Increasing – treads, spikes, grips. Diffusion – movement of molecules from high concentration to low. Dilution – making a solution weaker.	pp. 126-128
8	1	**CROP GROWING**	Root tubers – cassava,	Crop – any plant grown and	pp. 128-130

			sweet potatoes	cared for. Selecting seeds and garden site. Root tubers – cassava, sweet potatoes – propagation and harvesting.	
	2		Root tubers continued – carrots, turnips; stem tubers	Carrots and turnips – propagation by seeds, harvesting. Stem tubers – Irish potatoes, white yams. Propagation by stem tubers; harvesting. Contrast with root tubers.	pp. 130-132
	3		Caring for tuber crops	Practices – row planting (advantages), thinning, weeding, pruning, watering, pest and disease control.	pp. 132-133
	4		Plant support and pests of tuber crops	Support – staking, trellising, propping. Advantages. Crop pests – meaning, vermin. Common pests for cassava, sweet potatoes, carrots, Irish potatoes. Signs of pest damage.	pp. 133-135
	5		Diseases of tuber crops	Common diseases – cassava mosaic, bacterial blight, wilt, leaf spot, potato blight, etc. Signs of disease damage. Control – spraying, crop rotation, resistant varieties, burning infected plants.	pp. 135-136
9	1		Storage of tuber crops	Methods – granaries, silos, stores, sacks on raised stands. Characteristics of a good store – dry, ventilated, rat-proof. Drying crops before storage. Rat guards.	pp. 136-137
	2		Science oriented clubs	Science oriented clubs –Young Farmers' Club, First Aid Club, Science and Technology Club, etc. Importance of science clubs to learners.	pp. 137-138
	3		Co-operative societies	Co-operative society – meaning, importance to farmers (providing quality seeds, tools, loans, organising tours, teaching better farming methods).	p. 138
	4	**BACTERIA AND FUNGI**	Bacteria – definition, reproduction and where found	Bacteria – tiny organisms made of a single cell. Reproduction by binary fission. Places found – air, water, soil, rotting matter. Aerobic and anaerobic bacteria. Movement by flagellum.	pp. 139-140
	5		Shapes of bacteria	Rod-shaped (bacilli) – TB; spherical (cocci) – pneumonia; spiral – syphilis; comma –	pp. 140-141

				cholera. Grouping bacteria by shape, locomotion, and way of feeding.	
10	1		Useful bacteria	Examples of useful bacteria – nitrogen-fixing, putrefying, Lactococcus, Bacillus coagulans. Uses – humus formation, nitrogen fixation, yoghurt/cheese making, biogas, vaccines, digestion.	pp. 141-142
	2		Harmful bacteria and control	Harmful bacteria – disease, food spoilage, wound infection. Prevention and control – antibiotics, antiseptics, disinfectants, sterilisation, proper hygiene, immunisation.	pp. 142-143
	3		Fungi – definition, examples and mushroom structure	Fungi – living things reproducing by spores, feed saprophytically. Examples – mushroom, toadstools, moulds, yeast. Mushroom structure – cap, gills, stipe, mycelium – functions.	pp. 143-145
	4		Useful fungi	Uses – food (mushrooms), medicine (penicillin from Penicillium), baking and brewing (yeast), humus formation.	pp. 145-146
	5		Harmful fungi and control	Human fungal diseases – ringworm, athlete's foot, candidiasis. Plant diseases – Panama, blight, leaf rust. Prevention – hygiene, pickling, fungicides, burning infected plants.	p. 146
11	1		Comparison of bacteria and fungi	Similarities – both cause diseases, break down organic matter, can be useful. Differences – size, reproduction (binary fission vs spores/budding), cell number.	pp. 146-147
	2		Famous scientists	Alexander Fleming – penicillin. Louis Pasteur – pasteurisation. Robert Koch – TB cause. Edward Jenner – smallpox vaccine. Joseph Lister – antiseptics. Sir Ronald Ross – malaria.	pp. 147-148
	3		**TOPIC EXTENSION – SOIL**	Further discussion on soil conservation and fertility improvement from the notes (compost details, mulching benefits, crop rotation patterns).	pp. 91-93, 96-97
	4		**TOPIC EXTENSION –**	Further exploration of heat	pp.

			MATTER AND ENERGY	transfer applications (thermos flask details, ventilation, land/sea breezes), and friction in daily life.	111-114, 126-127
	5		**TOPIC EXTENSION – CROP GROWING AND MICROBES**	Deeper look at pest and disease control methods, integrated pest management from notes, and the role of microorganisms in soil fertility.	pp. 135-136, 141-142
12	1–5	**TERM II REVISION**	Comprehensive revision of all Term II topics: Soil, Matter and Energy, Crop Growing, Bacteria and Fungi. Past-paper style questions, oral drills, diagram labelling, group discussions		

LESSON PLAN 1: WEEK 1 PERIOD 1

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: SOIL FORMATION AND TYPES (pp. 81-83)

CONTENT:

• Soil is the top most layer of the earth's surface.

• Soil formation occurs by weathering (the process by which rocks are broken down into small particles to form soil) and decomposition.

• Causes of weathering: temperature changes, action of water, animal movement, plant roots.

• Types of soil: loam soil, sand soil, clay soil.

COMPETENCES (The learner;):

• Defines soil as the top most layer of the earth's surface.

• Names the two processes by which soil is formed.

• Lists at least three causes of weathering.

• Names the three types of soil.

• Draws simple diagrams showing particle arrangement in each type of soil.

• Differentiates between weathering and decomposition.

METHODS:

• Direct instruction

• Diagram study

• Question and answer

• Note-taking

LIFE SKILLS / VALUES:

• Appreciation of natural resources

• Observation skills

TEACHING/LEARNING MATERIALS:

- Samples of loam, sand, and clay soil in transparent containers
- Diagrams showing particle arrangement (from pp. 83-84)
- Chalkboard
- Pupils' exercise books
- Reference: pp. 81-83

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Asks: "What is the ground we walk on made of?" • Displays three soil samples (loam, sand, clay). Asks: "What differences do you see in these soils?" • Tells learners they will learn how soil is formed and the different types.	Question and answer Observation	• Observe the soil samples. • Share observations about colour and texture.
15 mins	EXPLORE	• Asks learners to look closely at each soil sample. • Asks: "How do you think rocks turn into tiny soil particles? What happens to dead plants and animals on the ground?" • Introduces the terms 'weathering' and 'decomposition'.	Discussion Question and answer	• Suggest how rocks break down. • Share ideas about rotting of dead matter.
20 mins	EXPLAIN	• Defines soil as the top most layer of the earth's surface. • Explains that soil is formed by weathering and decomposition. • Lists causes of weathering: temperature changes, action of water, animal movement, plant roots. • Names the three soil types based on particle size and arrangement. • Instructs learners to copy notes and draw simple particle arrangements from the textbook diagrams.	Direct instruction Note-taking Drawing	• Copy notes. • Draw particle arrangements for sand, loam, clay. • Ask questions.
10 mins	ELABORATE	• Re-explains: "Weathering is the process by which rocks are broken down into small particles to form soil. Decomposition is the rotting of dead plants and animals." • Writes an activity on the board: "1. What is soil? 2. Name the two ways by which soil is formed. 3. Give any two causes of weathering." • Instructs learners to copy and complete.	Direct elaboration Board activity	• Copy and answer the questions.
5	EVALUATE	• Asks selected learners to read their	Oral questioning	• Read answers.

mins		answers. • Quick quiz: "What is weathering? Name one cause. What are the three types of soil?"		• Answer quiz questions.
5 mins	EXTEND	• Home assignment: "At home, collect a small sample of soil from your compound. Describe its colour and feel. Write one sentence about your sample." • Next lesson: **CHARACTERISTICS AND USES OF SOIL TYPES.**	Home assignment Topic preview	• Note the assignment. • Listen for next lesson title.

ACTIVITY

1. What is soil?
2. Name the two ways by which soil is formed.
3. Give any two causes of weathering.

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 2: WEEK 1 PERIOD 2

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: CHARACTERISTICS AND USES OF SOIL TYPES (pp. 83-84)

CONTENT:

• Loam soil – has a lot of humus, is well aerated, has balanced particles of clay and sand soil, particles are fairly arranged. Used for growing crops.

• Sand soil – has large particles, large air spaces, dries up quickly, does not contain humus, has the highest rate of drainage, has the lowest rate of capillarity. Used for making glasses, used in building.

• Clay soil – has fine particles, tiny air spaces, closely packed particles, little humus, retains water for a long time, has the highest capillarity, is sticky. Used for making pots, cups and plates, used for making bricks, used for making roofing tiles.

COMPETENCES (The learner;):

• States at least two characteristics of each soil type.
• Gives at least one use of each soil type.
• Explains why loam soil is best for growing crops.
• Explains why clay soil is used in modelling.
• Draws a table comparing the three soil types.
• States which soil has the highest and lowest drainage rate.

METHODS:

• Direct instruction
• Observation and recording
• Question and answer
• Note-taking

LIFE SKILLS / VALUES:

• Appreciation of soil resources
• Practical knowledge application

TEACHING/LEARNING MATERIALS:

• Samples of loam, sand, clay
• Diagrams of particle arrangement
• Chalkboard
• Pupils' exercise books
• Reference: pp. 83-84

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews: "What is soil? Name the two ways soil is formed. Name the three types of soil." • Asks: "Which soil do farmers like most? Which one do builders use? Today we learn the characteristics and uses."	Review Question and answer	• Answer review questions. • Share what they know.
15 mins	EXPLORE	• Distributes small samples of the three soils. Asks learners to feel and describe the texture. • Asks: "Which one feels gritty? Which one feels sticky? Which one is dark and crumbly?" • Discusses how the feel relates to water movement and holding.	Hands-on observation Discussion	• Feel the soil samples. • Describe textures. • Link texture to water behaviour.
20 mins	EXPLAIN	• Writes characteristics of each soil type exactly from the notes: Loam – has a lot of humus, well aerated, balanced particles; Sand – large particles, large air spaces, dries quickly, no humus, highest drainage, lowest capillarity; Clay – fine particles, tiny air spaces, closely packed, little humus, retains water, highest capillarity, sticky. • Lists uses exactly from notes. • Instructs learners to copy notes in a table format.	Direct instruction Note-taking	• Copy the table. • Ask questions.
10	ELABORATE	• Re-explains: "Loam is the best for	Direct elaboration	• Copy and

mins		farming because it has humus and balanced particles that hold air and water. Clay retains water longest because of tiny spaces, so it's good for making pots that hold water. Sand drains quickly." • Writes activity: "Copy and complete the table: Soil type – Two characteristics – One use (for loam, sand, clay)."	Board activity	complete the table.
5 mins	EVALUATE	• Quick quiz: "Which soil is best for growing crops? Why? Which soil is used for making pots? Why does sand soil dry up quickly?"	Oral questioning	• Answer quiz.
5 mins	EXTEND	• Home assignment: "At home, find one object made from clay and one made using sand. Write their names and explain why that soil type was used." • Next lesson: **SOIL DRAINAGE.**	Home assignment Topic preview	• Note assignment. • Listen for next lesson title.

ACTIVITY

Copy and complete the table:

Soil type	Two characteristics	One use
Loam		
Sand		
Clay		

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 3: WEEK 1 PERIOD 3

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: SOIL DRAINAGE (pp. 84-85)

CONTENT:

• Soil drainage is the downward movement of water in the soil.

• OR: The capacity of the soil to allow water pass through it.

• Experiment to show soil drainage: equipment – samples of sand, loam and clay soil, empty bottle, cotton wool, funnel.

• Steps: (i) get samples of dry sand, clay, loam soil and three funnels, three bottles, cotton wool; (ii) put some cotton wool in each funnel; (iii) put each type of soil in separate funnel; (iv) place each funnel in a bottle and pour equal

amount of water into the soil; (v) observe what happens as water flows through each type; (vi) measure the amount of water collected in each bottle using a measuring cylinder.
• Results: much water collects from sand (highest drainage), little water collects from clay (lowest drainage), moderate from loam. Sand has largest air spaces, clay has tiniest air spaces.

COMPETENCES (The learner;):
• Defines soil drainage as the downward movement of water in the soil.
• Describes the experiment to show soil drainage.
• States which soil has the highest and lowest drainage rate.
• Explains why sand soil drains faster than clay soil.
• Draws and labels the experimental setup.
• Predicts outcomes of a similar drainage experiment.

METHODS:
• Direct instruction
• Demonstration (or diagram)
• Question and answer
• Note-taking

LIFE SKILLS / VALUES:
• Scientific inquiry
• Observation and interpretation

TEACHING/LEARNING MATERIALS:
• Funnels, cotton wool, three plastic bottles, sand, loam, clay, water, measuring cylinder (or clear diagrams)
• Chalkboard
• Pupils' exercise books
• Reference: pp. 84-85

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews soil types: "Which soil has the largest particles? Which has the smallest?" • Asks: "If you pour water onto sand and onto clay, which will let water pass through faster? Today we measure this – it is called drainage."	Review Question and answer	• Answer review. • Predict outcomes.
15 mins	EXPLORE	• Sets up the drainage experiment with three funnels and bottles (or shows diagram). • Asks learners to predict which bottle will collect the most water. • Pours equal water into each funnel. Asks learners to observe and record the amount of water collected.	Demonstration Prediction	• Predict. • Observe and record results.
20	EXPLAIN	• Defines soil drainage exactly: "Soil	Direct instruction	• Copy notes.

mins		drainage is the downward movement of water in the soil." • Describes the procedure step by step as in the notes. • Writes results: sand – highest drainage; clay – lowest; loam – moderate. • Explains: sand has large air spaces → water passes quickly; clay has tiny air spaces → water passes slowly. • Instructs learners to copy notes and draw the setup.	Note-taking Drawing	• Draw and label the experimental setup. • Ask questions.
10 mins	ELABORATE	• Re-explains: "Drainage is important – if water stays too long, roots rot; if it drains too fast, plants dry out. Loam is balanced." • Writes activity: "Complete: 1. Soil drainage is ____. 2. The soil with the highest drainage rate is ____. 3. The soil with the lowest drainage rate is ____ because ____."	Direct elaboration Board activity	• Copy and complete.
5 mins	EVALUATE	• Quick quiz: "Which soil has the highest drainage rate? Give a reason why clay soil retains water for a long time."	Oral questioning	• Answer quiz.
5 mins	EXTEND	• Home assignment: "At home, with an adult's help, pour water into two different soils in the garden. Observe what happens. Write two sentences about your observation." • Next lesson: **SOIL CAPILLARITY.**	Home assignment Topic preview	• Note assignment. • Listen for next lesson title.

ACTIVITY

Copy and complete:

1. Soil drainage is _____.
2. The soil with the highest drainage rate is _____.
3. The soil with the lowest drainage rate is _____ because _____.

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 4: WEEK 1 PERIOD 4

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: SOIL CAPILLARITY (pp. 85-86)

CONTENT:

• Soil capillarity is the upward movement of water in the soil.

• Experiment to find out the capillarity of different soils:

– Fill three long glass tubes with ends open with sand, clay and loam soils.

– Plug the lower end of each glass tube with some cotton wool.

– Put the glass tubes in a trough with water.

– After few hours the water level rises through the soils in each glass tube.

• Results: water rises highest in clay soil (highest capillarity), lowest in sand soil (lowest capillarity), moderate in loam.

• Explanation: clay soil has tiny air spaces → stronger capillary pull; sand soil has large air spaces → weak capillary action.

COMPETENCES (The learner;):

• Defines soil capillarity as the upward movement of water in the soil.

• Describes the experiment to compare capillarity of different soils.

• States which soil has the highest and lowest capillarity.

• Explains why clay soil shows the highest capillarity.

• Draws and labels the experimental setup.

• Predicts the capillary rise in different soils.

METHODS:

• Direct instruction

• Demonstration (or diagram)

• Question and answer

• Note-taking

LIFE SKILLS / VALUES:

• Curiosity about natural phenomena

• Application of scientific principles

TEACHING/LEARNING MATERIALS:

• Three long glass tubes (or transparent straws), cotton wool, sand, loam, clay, a trough of water (or clear diagram of the setup)

• Chalkboard

• Pupils' exercise books

• Reference: pp. 85-86

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews drainage: "Which soil allowed water through fastest? What might cause water to move upward in	Review Demonstration Question and answer	• Observe paper towel. • Suggest why

		soil?" • Shows a paper towel dipped in water; asks: "Why does water climb up the paper? Today we learn about capillarity in soil."		water moves up.
15 mins	EXPLORE	• Sets up the capillarity experiment (or shows diagram). • Asks learners to predict: "In which soil will water rise highest? Lowest? Why?" • After some time (or using prepared results), shows the water levels. • Asks learners to compare results with predictions.	Demonstration Prediction	• Make predictions. • Observe results. • Compare predictions to results.
20 mins	EXPLAIN	• Defines soil capillarity exactly: "Soil capillarity is the upward movement of water in the soil." • Explains the experiment step by step as in the notes. • States results: clay – highest capillarity; sand – lowest; loam – moderate. • Gives reason: clay has tiny air spaces → stronger capillary pull; sand has large air spaces → weak capillary action. • Instructs learners to copy notes and draw the setup.	Direct instruction Note-taking Drawing	• Copy notes. • Draw and label the experimental setup. • Ask questions.
10 mins	ELABORATE	• Re-explains: "Capillarity helps plants get water from deep in the soil to their roots. Clay holds water strongly because of tiny spaces, so plants can get moisture even when it hasn't rained." • Writes activity: "Complete: 1. Soil capillarity is ____. 2. The soil with the highest capillarity is ____ because ____. 3. The soil with the lowest capillarity is ____."	Direct elaboration Board activity	• Copy and complete.
5 mins	EVALUATE	• Quick quiz: "Which soil has the highest capillarity? Why? What property helps water move from deep soil to the surface?"	Oral questioning	• Answer quiz.
5 mins	EXTEND	• Home assignment: "At home, place a piece of chalk standing in a little water in a plate. Observe what happens after 10 minutes and write a short explanation using the term 'capillarity'." • Next lesson: **COMPONENTS OF**	Home assignment Topic preview	• Note assignment. • Listen for next lesson title.

		SOIL.		

ACTIVITY

Copy and complete:

1. Soil capillarity is _____.
2. The soil with the highest capillarity is _____ because _____.
3. The soil with the lowest capillarity is _____.

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 5: WEEK 1 PERIOD 5

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: COMPONENTS OF SOIL (pp. 86-88)

CONTENT:

• Components of soil: humus, air, water, living things/organisms, rock particles, dissolved mineral salts.

• Humus – formed by decomposition of plant and animal matter; makes soil fertile.

• Air – found in spaces between particles; used by seeds during germination, by plant roots for respiration, by living things for respiration. Experiment: when dry soil is put in water, air bubbles come out showing soil contains air.

• Water – used by plants during germination and photosynthesis; dissolves mineral salts. Experiment: heating soil in a container shows water droplets on the lid proving soil contains water.

• Living organisms – earthworms, bacteria, fungi, termites, mole rats. They aerate the soil (create tunnels); bacteria speed up decay to form humus.

• Rock particles – formed as a result of weathering; contain mineral salts necessary for plant growth; create space for air.

• Dissolved mineral salts – iron (formation of chlorophyll), magnesium (formation of chlorophyll), phosphorus (strong plant cell wall), calcium (strengthens stems and leaves), potassium (resistance to diseases and drought), nitrates and phosphates (make plant proteins). Plant roots absorb mineral salts by osmosis.

COMPETENCES (The learner;):

• Lists at least four components of soil.

• Describes experiments to show soil contains air and water.

• States the importance of humus in soil.
• Names two living organisms found in soil and explains how they are useful.
• Lists at least three important mineral salts found in soil and their functions.
• Explains how plant roots absorb mineral salts from the soil.

METHODS:
• Direct instruction
• Demonstration (or diagrams of experiments)
• Question and answer
• Note-taking

LIFE SKILLS / VALUES:
• Understanding ecosystem components
• Appreciation of soil life

TEACHING/LEARNING MATERIALS:
• Dry soil sample, transparent container of water (for air experiment), tin lid/dry soil and heat source (or diagram for water experiment)
• Chalkboard
• Pupils' exercise books
• Reference: pp. 86-88

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews capillarity: "Which soil holds water best? What else is in soil besides water and rock particles?" • Drops dry soil into a jar of water – bubbles rise. Asks: "What are these bubbles? Where did they come from?" • Tells learners soil contains many components.	Review Demonstration Question and answer	• Observe bubbles. • Answer: air.
15 mins	EXPLORE	• Performs the water experiment setup (heating soil in a container with a lid) or shows diagram. • Asks learners to predict what will appear on the lid. • Discusses what humus is and why top soil is dark. • Asks: "What living things have you seen in soil?" (worms, ants).	Demonstration Discussion	• Predict water droplets. • Share observations about soil life.
20 mins	EXPLAIN	• Lists the components of soil exactly: humus, air, water, living things/organisms, rock particles, dissolved mineral salts.	Direct instruction Note-taking	• Copy notes. • Ask questions.

		• For each, gives exact definitions and importances from notes. • Describes the air and water experiments. • Lists mineral salts and their roles: iron and magnesium for chlorophyll, phosphorus for strong cell walls, calcium for stems, potassium for disease resistance, nitrates and phosphates for proteins. Roots absorb by osmosis. • Instructs learners to copy notes.		
10 mins	ELABORATE	• Re-explains: "Humus is the dark part from dead things; it makes plants grow well. Worms and bacteria are good for soil; they help break down dead things and make tunnels that let in air." • Writes activity on board: "List any four components of soil. Next to each, write one importance."	Direct elaboration Board activity	• Copy and complete the list.
5 mins	EVALUATE	• Quick quiz: "What component makes soil dark? How do we prove soil contains air? Name one mineral salt important for chlorophyll formation."	Oral questioning	• Answer quiz.
5 mins	EXTEND	• Home assignment: "At home, take a handful of soil and spread it on a white paper. Look closely. List all the different things you see (particles, tiny stones, dead plant bits). Bring your list tomorrow." • Next week: **SOIL PROFILE AND SOIL EROSION.**	Home assignment Topic preview	• Note assignment. • Listen for next week's topic.

ACTIVITY

List any four components of soil. Next to each component, write one importance.

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 6: WEEK 2 PERIOD 1

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: SOIL PROFILE (p. 88)

CONTENT:

• Soil profile is the arrangement of soil layers from top to bottom.

• OR: Is the vertical arrangement of soil layers.

• Layers:

– Top soil: uppermost layer, dark in colour, has a lot of humus, good for growing crops, contains many living organisms.

– Sub soil: second layer, lighter in colour, less humus, contains some minerals and rock fragments.

– Parent rock / weathered rock: lowest layer, partly broken rock from which soil is formed.

• Soil profile can be clearly seen in pit latrines and trenches.

COMPETENCES (The learner;):

• Defines soil profile as the arrangement of soil layers from top to bottom.

• Names the three main layers of a soil profile.

• States the characteristics of each layer.

• Draws and labels a diagram of a soil profile.

• Explains why top soil is best for growing crops.

• Identifies places where a soil profile can be clearly seen.

METHODS:

• Direct instruction

• Diagrammatic presentation

• Question and answer

• Note-taking

LIFE SKILLS / VALUES:

• Observational skills

• Appreciation of earth structure

TEACHING/LEARNING MATERIALS:

• Large diagram of a soil profile showing the three layers

• Chalkboard

• Pupils' exercise books

• Reference: p. 88

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews components: "Name one component that makes soil fertile." • Asks: "If you dig a deep hole, does the soil look the same all the way down? What changes do you notice? Today we learn about the layers of soil – the soil profile."	Review Question and answer	• Answer: humus. • Share what they have noticed when digging.
15 mins	EXPLORE	• Displays a large cross-section diagram of a soil profile. • Asks learners to describe the colour differences from top to bottom.	Diagram study Discussion	• Observe the diagram. • Suggest answers.

		• Asks: "Which layer looks darkest? Why do you think? Where would you find most worms and insects?"		
20 mins	EXPLAIN	• Defines soil profile exactly: "Soil profile is the arrangement of soil layers from top to bottom." • Describes each layer with exact characteristics: top soil – dark, humus, fertile, many living organisms; sub soil – lighter, less humus, some minerals; parent rock – broken rock. • Names places where soil profile can be seen: pit latrines, trenches. • Instructs learners to copy notes and draw the soil profile, labelling the three layers.	Direct instruction Note-taking Drawing	• Copy notes. • Draw and label the soil profile diagram. • Ask questions.
10 mins	ELABORATE	• Re-explains: "Top soil is where farmers plant because it has nutrients from humus. The sub soil stores water and minerals that roots can reach. Parent rock is the source material that slowly breaks to make new soil." • Writes activity on board: "Draw a soil profile and label the three layers. Next to each layer, write one characteristic."	Direct elaboration Board activity	• Draw and complete labelling with characteristics.
5 mins	EVALUATE	• Asks a few learners to show their diagrams. • Quick quiz: "Which layer has the most humus? Where would you find parent rock? Why is top soil important for farmers?"	Oral questioning	• Show diagrams. • Answer quiz.
5 mins	EXTEND	• Home assignment: "At home, find a deep hole, road cut, or trench near your home (with an adult). Observe the layers you see. Draw what you see and label the layers using today's lesson." • Next lesson: **SOIL EROSION – MEANING AND CAUSES.**	Home assignment Topic preview	• Note assignment. • Listen for next lesson title.

ACTIVITY

Draw and label a soil profile showing the three main layers. Next to each layer, write one characteristic.

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 7: WEEK 2 PERIOD 2

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: SOIL EROSION – MEANING AND CAUSES (pp. 89-90)

CONTENT:

- Soil erosion is the removal of top soil by its agents.
- Agents of soil erosion – fast running water, strong wind, moving animals.
- Causes of soil erosion – bush burning, deforestation, overgrazing, overstocking, monocropping.
- Danger of soil erosion to crops – it makes soil unsuitable for plant growth.

COMPETENCES (The learner;):

- Defines soil erosion as the removal of top soil by its agents.
- Names the three main agents of soil erosion.
- Lists at least four causes of soil erosion.
- Explains how each cause leads to erosion.
- States one danger of soil erosion to crops.
- Suggests an area near the school that might be affected by erosion and explains why.

METHODS:

- Direct instruction
- Observation of pictures/diagrams
- Question and answer
- Note-taking

LIFE SKILLS / VALUES:

- Environmental awareness
- Responsibility for land care

TEACHING/LEARNING MATERIALS:

- Pictures or diagrams showing eroded land, deforestation, overgrazing
- Chalkboard
- Pupils' exercise books
- Reference: pp. 89-90

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews soil profile: "Which layer is best for growing crops? Why?" • Shows a picture of land with deep gullies and asks: "What happened to the soil here? Where did it go? Today we learn about soil erosion."	Review Picture observation Question and answer	• Answer review. • Observe the picture and suggest what happened.

15 mins	EXPLORE	• Asks: "What do you think can carry soil away? Have you seen soil being moved by water or wind?" • Shows pictures of a burnt field, a bare hill, animals grazing closely. Asks: "How do these activities cause soil to be removed?" • Introduces the term 'agents of erosion'.	Discussion Picture study	• Share observations. • Link pictures to causes.
20 mins	EXPLAIN	• Defines soil erosion exactly: "Soil erosion is the removal of top soil by its agents." • Lists agents: fast running water, strong wind, moving animals. • Lists causes: bush burning, deforestation, overgrazing, overstocking, monocropping. • Explains how each leads to erosion (vegetation removed, soil left bare). • States danger: makes soil unsuitable for plant growth. • Instructs learners to copy notes.	Direct instruction Note-taking	• Copy notes. • Ask questions.
10 mins	ELABORATE	• Re-explains: "When trees are cut and grass is eaten bare, there are no roots to hold the soil. Rainwater or wind easily carries the top soil away. This is erosion." • Writes activity on board: "List the three agents of soil erosion. For each, write one cause of erosion linked to that agent."	Direct elaboration Board activity	• Copy and complete the activity.
5 mins	EVALUATE	• Asks learners to read their answers. • Quick quiz: "Name two agents of soil erosion. Why is bush burning a cause of soil erosion?"	Oral questioning	• Read answers. • Answer quiz.
5 mins	EXTEND	• Home assignment: "Look around your home or on the way to school for signs of soil erosion. Write a short description of what you saw (e.g., gullies, bare soil). Bring your report tomorrow." • Next lesson: **TYPES OF SOIL EROSION AND EFFECTS.**	Home assignment Topic preview	• Note assignment. • Listen for next lesson title.

ACTIVITY

List the three agents of soil erosion. For each agent, write one cause of erosion that is linked to that agent.

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 8: WEEK 2 PERIOD 3

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: TYPES OF SOIL EROSION AND EFFECTS (pp. 90-91)

CONTENT:

• Types of soil erosion:

– Splash erosion – raindrops fall on bare land and splash soil particles away.

– Sheet erosion – top soil uniformly carried away by running water or wind.

– Rill erosion – shallow channels are created on the land by slow moving water.

– Gully erosion – big channels are created on the land by fast running water.

– River or bank erosion – this type of soil erosion occurs near river bank.

– Wind erosion – soil carried away by wind.

• Effects of soil erosion: leads to soil exhaustion; leads to poor yields in crops.

• Silting – the deposition of silt or eroded materials into the water body. Causes of silting: clearing vegetation around river banks, deforestation, bush burning, cultivating near river banks. Effects of silting: makes water body shallow, leads to death of aquatic life, leads to water pollution. Ways of controlling silting: avoiding clearing vegetation around river banks, avoiding cultivating near river banks, afforestation, avoiding bush burning.

COMPETENCES (The learner;):

• Names and describes at least four types of soil erosion.

• Distinguishes between rill and gully erosion.

• States two effects of soil erosion.

• Defines silting and explains its effects.

• Draws simple sketches showing rill and gully erosion.

• Identifies which type of erosion might occur in a given scenario.

METHODS:

• Direct instruction

• Diagrammatic illustrations

• Question and answer

• Note-taking

LIFE SKILLS / VALUES:

• Environmental conservation

• Critical observation

TEACHING/LEARNING MATERIALS:

• Diagrams or pictures of each erosion type

• Chalkboard

• Pupils' exercise books
• Reference: pp. 90-91

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews agents and causes: "Name one agent of soil erosion. What happens when there are no trees on a hill?" • Shows a picture of a deep gully and asks: "What type of erosion made this?"	Review Question and answer	• Answer review. • Observe picture.
15 mins	EXPLORE	• Displays diagrams of splash, sheet, rill, gully, river bank, and wind erosion. • Asks: "Which one looks like tiny channels? Which one is a big, deep cut in the land? Which one happens near a river?" • Discusses the differences.	Diagram study Class discussion	• Study diagrams. • Identify each type from description.
20 mins	EXPLAIN	• Defines each type of erosion exactly as in notes with examples. • Explains effects: soil exhaustion, poor yields. • Defines silting: the deposition of silt or eroded materials into the water body. Lists causes, effects, and ways of controlling silting. • Instructs learners to copy notes and draw simple diagrams of rill and gully erosion.	Direct instruction Note-taking Drawing	• Copy notes. • Draw rill and gully erosion. • Ask questions.
10 mins	ELABORATE	• Re-explains: "Sheet erosion is like a thin blanket of soil being washed away. Rills are tiny paths; gullies are deep cuts that cannot be crossed easily. Silting damages fish habitats." • Writes activity on board: "Match the erosion type to its description: Splash / Sheet / Rill / Gully — (a) uniform removal of top soil; (b) deep channels; (c) raindrop impact; (d) shallow channels."	Direct elaboration Board activity	• Copy and match.
5 mins	EVALUATE	• Quick quiz: "What is the difference between rill and gully erosion? What is silting? Give one effect of silting."	Oral questioning	• Answer quiz.
5 mins	EXTEND	• Home assignment: "Draw a picture of a landscape showing at least two different types of erosion. Label them."	Home assignment Topic preview	• Note assignment. • Listen for next lesson title.

		• Next lesson: **CONTROL OF SOIL EROSION.**		

ACTIVITY

Match the erosion type to its description:

(a) Splash erosion – (i) deep channels cut by fast water

(b) Sheet erosion – (ii) raindrop impact on bare soil

(c) Rill erosion – (iii) uniform removal of top soil

(d) Gully erosion – (iv) shallow channels by slow water

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 9: WEEK 2 PERIOD 4

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: CONTROL OF SOIL EROSION (pp. 91-93)

CONTENT:

• Methods of controlling soil erosion: mulching, crop rotation, terracing, contour ploughing, strip cropping, afforestation, re-afforestation, bush fallowing, agro-forestry.

• Mulching is the covering of soil using dry plant materials. Advantages: keeps moisture in the soil, controls soil erosion, improves soil fertility.

• Crop rotation is the growing of different crops on the same piece of land seasonally. Advantages: controls soil erosion, controls crop pests and diseases, improves soil fertility.

• Terracing is the digging of terraces on a slope or hilly area. Terraces slow down the speed of running water.

• Strip cropping is the growing of crops on strips across the slope.

• Contour ploughing is the ploughing of land across the slope.

• Soil leaching is the loss of soil nutrients from the upper layers to the deeper layers where plant roots cannot reach.

• Soil exhaustion is the loss of soil fertility.

COMPETENCES (The learner;):

• Lists at least five methods of controlling soil erosion.

• Defines mulching, crop rotation, and terracing.

• Describes how mulching, crop rotation, and terracing control erosion.

• Explains why mulching is done in wet season.

• Defines soil leaching and soil exhaustion.

• Suggests suitable erosion control methods for a hilly school compound.

METHODS:

• Direct instruction

• Diagrammatic explanation

• Question and answer
• Note-taking

LIFE SKILLS / VALUES:
• Environmental stewardship
• Practical farming knowledge

TEACHING/LEARNING MATERIALS:
• Diagrams of terracing, contour ploughing, strip cropping
• Samples of mulch materials (dry grass, banana leaves)
• Chalkboard
• Pupils' exercise books
• Reference: pp. 91-93

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews types of erosion: "Which type creates deep channels? How can we stop this?" • Shows a piece of dry grass and asks: "If I spread this on soil, how might it protect the soil from rain?"	Review Question and answer	• Answer review. • Suggest how mulch protects soil.
15 mins	EXPLORE	• Shows diagrams of a terraced hillside, contour ploughed field, and strip cropped slope. • Asks: "What do you notice about how the land is shaped? How does this help stop water from running fast?" • Discusses the idea of slowing down water.	Diagram study Discussion	• Observe diagrams. • Explain how terracing slows water.
20 mins	EXPLAIN	• Defines each method exactly from notes: mulching, crop rotation, terracing, contour ploughing, strip cropping, afforestation, bush fallowing, agro-forestry. • Explains how each controls erosion. • Defines soil leaching and soil exhaustion. • Instructs learners to copy notes.	Direct instruction Note-taking	• Copy notes. • Ask questions.
10 mins	ELABORATE	• Re-explains: "Mulching is like a blanket for the soil. Terracing turns a hill into steps so water cannot run straight down. Crop rotation means changing crops each season so the soil is always covered." • Writes activity on board: "List three methods of controlling	Direct elaboration Board activity	• Copy and complete the activity.

		erosion. Next to each, write one sentence explaining how it works."		
5 mins	EVALUATE	• Quick quiz: "What is terracing? Name one method that involves planting trees. Why is mulching done in wet season?"	Oral questioning	• Answer quiz.
5 mins	EXTEND	• Home assignment: "Suggest two methods that could be used to control erosion on a steep slope near your home. Explain your choice." • Next lesson: **SOIL FERTILITY AND ORGANIC MANURE.**	Home assignment Topic preview	• Note assignment. • Listen for next lesson title.

ACTIVITY

List three methods of controlling soil erosion. Next to each, write one sentence explaining how it works.

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

LESSON PLAN 10: WEEK 2 PERIOD 5

DATE: _____ | **CLASS:** Primary Five | **NO. OF LEARNERS:** _____ | **SUBJECT:** Science | **TIME:** 60 Minutes

TOPIC: COMPONENTS OF THE ENVIRONMENT (SOIL)

SUB TOPIC: SOIL FERTILITY AND ORGANIC MANURE (pp. 93-97)

CONTENT:

• Soil fertility is the ability of the soil to support plant growth.

• Ways of improving soil fertility: using fertilizers, mulching, bush fallowing, agro-forestry, crop rotation.

• Fertilizers are substances put in the soil to increase its fertility. Types: natural fertilizers (organic manure) and artificial fertilizers (inorganic fertilizers).

• Organic manure / natural fertilizers: manure got from decayed plants and animal materials. Types: compost manure, farmyard manure, green manure, organic mulches.

• Compost manure is the type of manure got from decayed plants and animal wastes. Methods of making compost manure: pit method and heap method.

• Things used to make compost manure: banana peeling, plant leaves, left over food, maize stalks, cassava peeling.

• Steps: put hard materials at bottom to trap nutrients and allow air; add farmyard manure to provide organisms that cause decay; add ash and sawdust to improve nutrient content; add a layer of top soil to increase rotting rate; sprinkle water to speed up rotting; turn materials from time to time; use a stick to check decomposition process.

• Advantages of organic manure: rich source of humus, improves soil structure, cheap, stays longer in the soil, does not pollute the soil.

• Disadvantages: tiresome to make, produce bad smell, bulky, take long to replace nutrients.

COMPETENCES (The learner;):

• Defines soil fertility as the ability of the soil to support plant growth.
• Lists ways of improving soil fertility.
• Defines fertilizers and names the two types.
• Lists types of organic manure.
• Describes the steps in making compost manure using the pit method.
• States at least two advantages and two disadvantages of organic manure.

METHODS:

• Direct instruction
• Demonstration (compost materials or pictures)
• Question and answer
• Note-taking

LIFE SKILLS / VALUES:

• Practical farming skills
• Recycling organic waste

TEACHING/LEARNING MATERIALS:

• Chart showing compost-making steps
• Samples of compost materials (dry grass, peels)
• Chalkboard
• Pupils' exercise books
• Reference: pp. 93-97

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	• Reviews erosion control: "Name one method of controlling erosion." • Shows a healthy plant in rich dark soil and a weak plant in poor soil. Asks: "Why is one plant healthier? What makes soil good for plants?" • Introduces the term 'soil fertility'.	Review Observation Question and answer	• Answer review. • Observe plants. • Suggest "fertile soil".
15 mins	EXPLORE	• Shows materials used for making compost: banana peels, dry grass, maize stalks. • Asks: "What happens when these things rot? How can rotting materials help soil?" • Introduces the term 'organic manure'. • Asks if anyone has seen a compost pit at home.	Discussion Real-life connection	• Share experiences. • Predict how rotting helps soil.
20 mins	EXPLAIN	• Defines soil fertility exactly: "Soil fertility is the ability of the soil to support plant growth." • Lists ways of improving fertility.	Direct instruction Note-taking	• Copy notes. • Draw pit diagram. • Ask questions.

		• Defines fertilizers and names the two types. • Explains organic manure: compost, farmyard, green manure, organic mulches. • Explains compost making – materials, pit method steps as in notes. • Lists advantages and disadvantages of organic manure exactly from notes. • Instructs learners to copy notes and draw a labelled compost pit.		
10 mins	ELABORATE	• Re-explains: "Hard materials like stalks go at the bottom to trap nutrients and let in air. Water and manure speed up rotting. The stick is used to check how far decomposition has gone." • Writes activity on board: "List four materials used to make compost manure. State why a layer of top soil is added."	Direct elaboration Board activity	• Copy and complete the activity.
5 mins	EVALUATE	• Quick quiz: "What is soil fertility? Give one advantage and one disadvantage of organic manure. Why should compost materials be turned?"	Oral questioning	• Answer quiz.
5 mins	EXTEND	• Home assignment: "At home, collect the materials you would need to make a small compost heap. List them in your book. If possible, start a small compost pile with help from an adult." • Next week: **ARTIFICIAL FERTILIZERS, SOIL POLLUTANTS AND SOIL BARRIERS.**	Home assignment Topic preview	• Note assignment. • Listen for next week's topic.

ACTIVITY

1. List four materials used to make compost manure.
2. Why is a layer of top soil added to the compost pit?

LESSON EVALUATION

Strengths observed:

Areas needing improvement:

Way forward / Remedial action:

Teacher's signature: _____ Date: _____

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LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 12: WEEK 3 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 13: WEEK 3 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

State	Has definite shape?	Has definite volume?	Molecular arrangement (describe)	One example

LESSON PLAN 14: WEEK 3 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 15: WEEK 3 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 16: WEEK 4 PERIOD 1

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 17: WEEK 4 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 18: WEEK 4 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 19: WEEK 4 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS'

				ACTIVITY

LESSON PLAN 20: WEEK 4 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 21: WEEK 5 PERIOD 1

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 22:WEEK 5 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 23: WEEK 5 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 24: WEEK 5 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 25: WEEK 5 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 26: WEEK 6 PERIOD 1

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 27: WEEK 6 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 28: WEEK 6 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 29: WEEK 6 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

Crop	How it is propagated	How it is harvested

LESSON PLAN 30: WEEK 6 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 31: WEEK 7 PERIOD 1

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

Crop	Two common pests

LESSON PLAN 32: WEEK 7 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 33:WEEK 7 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 34: WEEK 7 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 35: WEEK 7 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 36: WEEK 8 PERIOD 1

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 37: WEEK 8 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 38: WEEK 8 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 39: WEEK 8 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 40: WEEK 8 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 41: WEEK 9 PERIOD 1

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 42: WEEK 9 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 43: WEEK 9 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 44: WEEK 9 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 45: WEEK 9 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 46: WEEK 10 PERIOD 1

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 47: WEEK 10 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 48: WEEK 10 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

Crop	One common pest	One common disease	One control method

LESSON PLAN 49: WEEK 10 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

Useful Bacteria	Role	Useful Fungi	Role

LESSON PLAN 50: WEEK 10 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 51: WEEK 11 PERIOD 1

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 52: WEEK 11 PERIOD 2

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS'

				ACTIVITY

LESSON PLAN 53: WEEK 11 PERIOD 3

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 54: WEEK 11 PERIOD 4

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

LESSON PLAN 55: WEEK 11 PERIOD 5

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY

ABOUT THIS BOOK

This comprehensive guide provides educators with structured, curriculum- aligned lesson plans designed to simplify teaching and improve learner performance. Each lesson includes clear objectives, teaching methods, activities and assessment strategies that save you time while ensuring quality delivery.